



04 — Domain Names



Introduction

After learning about websites, the client-server model, and web browsers, the next concept I want to understand is **domain names**.

Whenever I visit a website, I usually type a human-readable name such as:

```
google.com  
github.com  
youtube.com
```

Instead of typing an IP address.

Domain names make it easier for humans to access websites without having to remember numerical IP addresses.

1 What Is a Domain Name?

A **domain name** is a human-readable name used to identify a location or service on the Internet.

For example:

```
google.com
```

is a domain name.

Computers communicate across networks using IP addresses, but remembering an IP address for every website would be difficult.

Instead of:

```
142.250.x.x
```

I can use:

```
google.com
```

The **Domain Name System (DNS)** helps connect the domain name to the appropriate IP address.

Why Do We Need Domain Names?

Imagine having to remember an IP address for every website I visit.

Instead of:

```
Website A → 192.0.2.10  
Website B → 192.0.2.20  
Website C → 192.0.2.30
```

I can use:

```
Website A → website-a.com  
Website B → website-b.com  
Website C → website-c.com
```

Domain names make the Internet easier for humans to use.

3 Domain Name vs IP Address

A domain name and an IP address are not the same thing.

Domain Name

```
example.com
```

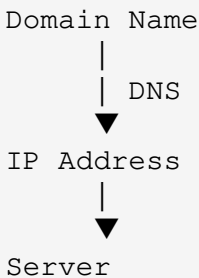
Human-friendly.

IP Address

```
93.184.216.34
```

Network address used to reach a destination.

A simplified relationship is:



Consider:

www.example.com

This can be broken down into different parts.

www.example.com

Top-Level Domain

Second-Level Domain

Subdomain

However, `www` is not actually required for a website to work.

For example:

example.com

www.example.com

can be configured to reach the same website or different services.

The final part of a domain name is called the **Top-Level Domain**, or TLD.

Examples include:

```
.com  
.org  
.net  
.edu  
.gov
```

There are also country-code TLDs:

```
.ke  
.uk  
.us  
.id  
.in
```

For example:

```
example.com  
  ↑  
  TLD
```

And:

```
example.ke  
  ↑  
  TLD
```

Second-Level Domain

Consider:

```
example.com
```

The word:

```
example
```

is the second-level domain.

Together:

```
example + .com
```

forms the domain name:

```
example.com
```

7 Subdomains

A domain can have subdomains.

For example:

```
blog.example.com
```

Here:

```
blog.example.com
|      |      |
|      |      | TLD
|      |      |
|      |      | Domain
|      |      |
|      |      | Subdomain
```

Common examples include:

```
www.example.com  
blog.example.com  
shop.example.com  
mail.example.com  
api.example.com
```

Different subdomains can be used for different services.

For example:

```
example.com  
|  
├── www.example.com  
├── blog.example.com  
├── shop.example.com  
└── api.example.com
```

Domain Name and Website

A domain name does not automatically mean there is a website behind it.

A domain can be used for different purposes, including:

- Websites
- Email
- APIs
- Applications
- Other Internet services

For example:

```
example.com
```

could point to a web server.

While:

```
mail.example.com
```

could point to a mail service.

9 Domain Registration

Before using a domain name, it generally needs to be registered through a **domain registrar**.

Examples of registrars include companies that allow people and organizations to register domain names.

The registration process can look like:

```
Choose Domain
  ↓
Check Availability
  ↓
Register Domain
  ↓
Configure DNS
  ↓
Connect Domain to Service
```

Domain registration does not necessarily mean that the website itself is hosted by the same company.

The domain and hosting can be managed separately.

10 Domain Registrar vs Web Host

This is an important distinction.

Domain Registrar

Provides domain registration services.

```
Registrar
  ↓
Domain Name
```

Web Host

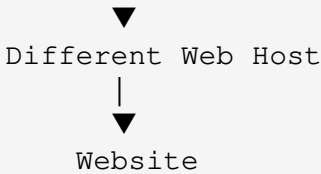
Provides infrastructure where website files or applications can run.

```
Web Host
  ↓
Server
  ↓
Website/Application
```

They can be the same company, but they don't have to be.

For example:

```
Domain Registrar
  |
  | example.com
  ▼
DNS
  |
```



🔍 What Happens When I Type a Domain?

Suppose I type:

```
https://example.com
```

into my browser.

A simplified process is:

1. I enter `example.com`
↓
2. Browser needs the IP address
↓
3. DNS lookup occurs
↓
4. DNS provides an IP address
↓
5. Browser connects to the server
↓
6. Browser sends an HTTP/HTTPS request
↓
7. Server responds
↓
8. Browser displays the website

This connects the topics I have already learned:

Domain Names



DNS



IP Address



Server



HTTP/HTTPS



Browser



Website

12 Domain Names and DNS

A domain name is closely connected to DNS.

DNS stands for:

Domain Name System

DNS translates names into information that computers can use to locate services.

For example:

example.com



DNS



IP Address



DNS is much more than simply converting domain names to IP addresses, and I will study it in greater detail in the next topic.

13 DNS Records

Domains can have different types of DNS records.

Some important records include:

Record	Purpose
A	Maps a domain to an IPv4 address
AAAA	Maps a domain to an IPv6 address
CNAME	Creates an alias for another domain name
MX	Specifies mail servers
TXT	Stores text information, often used for verification and email-related configurations
NS	Identifies authoritative name servers

For example:

```
example.com
├── A      → IPv4 address
├── AAAA   → IPv6 address
├── MX     → Mail server
└── TXT    → Verification information
```

I will explore these records more deeply when I study DNS.

14 Domain Name Example

Imagine I create a website for my IT business.

I might register:

```
mukhtaartech.com
```

I could then create subdomains:

```
mukhtaartech.com
|
├── www.mukhtaartech.com
├── support.mukhtaartech.com
├── portal.mukhtaartech.com
└── api.mukhtaartech.com
```

Each could potentially be configured for a different purpose.

For example:

```
portal.mukhtaartech.com
    ↓
Customer Portal

support.mukhtaartech.com
    ↓
Support System

api.mukhtaartech.com
    ↓
Backend API
```

This is an example of how domain names can become part of a larger web application architecture.

1.5 Domain Names Are Hierarchical

Domain names have a hierarchical structure.

For example:

```
www.example.com
```

can be viewed as:

```
  .  
  |  
com  
  |  
example  
  |  
www
```

At the top is the DNS root.

Below the root are top-level domains such as:

```
.com  
.org  
.net  
.ke
```

Below the TLD is the domain.

Below the domain can be subdomains.

100 Domain Names Are Not the Same as URLs

These terms are related but different.

Domain

```
example.com
```

URL

```
https://example.com/about/contact
```

The URL contains more information.

A simplified URL:

```
https://example.com/about
|         |         |
|         |         | Path
|         |         |
|         |         | Domain
|         |         |
|         |         | Protocol
```

So:

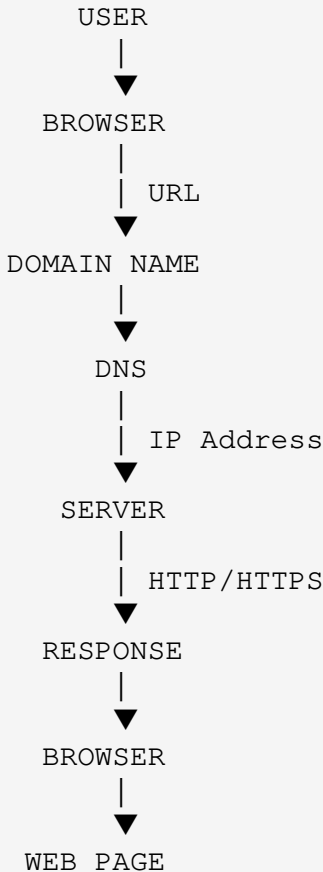
```
Domain = example.com
```

while:

```
URL = https://example.com/about
```

Putting Everything Together

The relationship between the concepts I have learned so far can be represented as:



What I Learned

From this topic, I learned that a domain name is a human-friendly way of identifying an Internet destination or service.

I learned that:

- Domain names are easier for humans to remember than IP addresses.
- DNS helps connect domain names with network information.
- A domain name is different from a URL.
- Domains can have subdomains.
- Domain registration and web hosting are separate concepts.
- DNS records can provide different types of information.
- Domains can be used for websites, email, APIs, and other services.
- A domain does not automatically mean that the website is hosted by the domain registrar.



Key Terms

Term	Meaning
Domain Name	Human-readable name used to identify an Internet service or destination
TLD	Top-Level Domain such as .com or .ke
Subdomain	A domain created under another domain
Registrar	Company/service through which domain names can be registered
DNS	System that helps resolve domain names and other DNS information
IP Address	Network address used to identify a destination
URL	Complete address used to locate a resource
DNS Record	Information stored in DNS for a domain

A Record	DNS record for an IPv4 address
AAAA Record	DNS record for an IPv6 address
CNAME	DNS alias pointing to another domain name
MX Record	Specifies mail servers
NS Record	Identifies authoritative name servers

? Questions I Want to Explore Next

- How does DNS actually work?
 - What happens during a DNS lookup?
 - What is a DNS resolver?
 - What are root name servers?
 - What are TLD name servers?
 - What are authoritative name servers?
 - How does DNS caching work?
 - What is DNS propagation?
 - How do I configure DNS records for my own domain?
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Progress

Stage: Web Fundamentals

Topic: Domain Names

Status:  Completed

Previous: [Web Browsers](#)

Next: DNS

Domain names give the Internet human-friendly names, while DNS helps turn those names into information computers can use.